

Campus Rent: A Full-Stack Web Application for Item Renting in University Environments

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Abstract— Campus Rent is a full-stack web platform that powers item rentals on university campuses in a peer-to-peer manner. It aims to replace fragmented owner-centrism with an access-based model, enabling students to rent electronics, tools, sports equipment, furniture, and more seamlessly on the portal. It combines a responsive frontend of React with a Node.js/Express- based backend and MongoDB as the database. Payment integration has been performed with the Razor pay gateway, which facilitates card-based, UPI-based, or cash-on-delivery-based transactions. The key functionalities of the site are user authentication; posting of items with categorization, real-time booking, and an admin dashboard for community moderation. It showed that rentals could be completed in under two minutes and handled hundreds of users at the same time with very little delay during testing. At the campus level, Campus Rent will advocate "rent-instead-of-owning" to ensure affordability and sustainability. In the future, further development could add functions such as blockchain contracts for convenience and trust-building.

Keywords— *Campus Rent, Item Renting, peer-to-peer rental, sharing economy, full-stack web development, campus sustainability*

I. INTRODUCTION

Today, people don't always want to own everything that they use. For many products, being able to borrow them for a while is akin to having a store that lends things out and takes them back. Peer-to-peer rental websites and apps are facilitating this shift by enabling those with extra stuff to connect with others who want it. That way, items such as tools, gadgets or sports gear no longer just gather dust but can be conveniently shared. This trend is based on the move toward access-based consumerism, and the fact that use when wanted is more important than ownership [1-2]. University students are a great example of those who can take advantage of access-based approaches. While they are away at school, there may be things that they need for the semester, such as electronics, lab equipment, sports gear or some basic furniture for their dorm [3].

Campus Rent is a student-focused solution to an all-too-common campus problem: not being able to get need, and instead buying it for exorbitant prices for such a short amount of time [4]. In each campus, there's a constant need for calculators, lab equipment, books, power banks, cycles, project kits, or other supplies for just a few days to spend in

every college or university. It's not as if these products are inexpensive to buy outright, either, and that amount is likely enough of a stretch for students trying to watch their pennies. In any case, many other students already possess these things; however, they have barely given orders to use them [5-6]. Campus Rent is bridging this gap by creating a safe online marketplace within the university through which students can lend and borrow to one another. It's an elegantly simple idea that leverages and shares resources with the campus, minimizes theft and nets savings. Since use of the site is limited only to those verifiable as real users and with a true student account, everyone on it shares being from the same institutional system, and can therefore warrant trust in one another at some higher level. In this way, not only does it operate as a marketplace of transactions, but it also promotes a culture of co-opetition and trustworthiness between students. Look at Campus Rent, for example, which does nothing more than turn college kids' garbage into a useful resource and one that can save students dollars [7].

The backend is 'multi-threaded', which means that author can serve more than one user at a time. The user profile, listed item, transaction record, payment information and the rating will be kept in a versatile document-oriented database system such as MongoDB [8-9]. It uses MongoDB for quick retrieval of data and would work for the same after adding the claims to the database. Strong authentication will be present so that user account protection is enforced. Overall, React & MongoDB make a great tech stack to keep the product fast, effective and would be relatively easy to maintain while developing new features [10-11]. The admin can monitor the new user accounts and check for suspicious use.

II. LITERATURE REVIEW

Recently Peer-to-peer digital enabled platforms, alongside web technology, mobile devices and cloud systems have facilitated recent growth in this area of modern rental operations. Previous studies widely covered rental scenario in limited domains such as transportation, housing and tourism. More recently, the focus of attention has switched to broader, more community focused rental models.

When online rental companies started, they generally offered only one type of service. Typically, a website would cater to either car bookings, property rentals or accommodation, but not all together. The author [11-12]

developed a housing rental portal in which users can book homes online, verify the profiles of one another, and communicate through their system. In their research, they concluded that the check-in process can feel safe and trustworthy only if it checks both owner and tenant information as well. Another author [13-14] proposed a car rental system with strengthened login and authentication procedures by which to decrease fake bookings as well as payment fraud. Development of a mobile application for a vehicle rental system. They wanted to make the app very easy to navigate and facilitate payments that are so quick that it feels simple and convenient. It eventually failed while it was limited to vehicles and did not address the broader shared items or other product domains [15]. The author [16] developed a full-stack car rental application contains a responsive page with integrated payment. They were able to show the efficiency of Node.js in handling high-volume parallel user interactions. This proves that full-stack approaches are fitting for high-demand rental platforms. However, all these systems lacked mixed-category item listings or verification tailored for a university-restricted user base. The researchers had also proposed a MERN-based system called "Flexible Sharing and Leasing" for vehicle rentals. They found that because Node.js uses a non-blocking structure, the platform can handle many users at the same time without slowing down [17-18].

The recent work focuses on multiple item-type rental platforms, rather than single-service ones. The author [19-20] proposed "Online Rental Things," a general-purpose website which allows users to create listings of products to be used, such as electronics, furniture, tools, and household items. Since it allows multiple categories, the system advances the share-and-borrow process of what one has and what one needs. Allowing the mentioned types of listings, the platform attempts to create trust among the users and motivate a sense of community and collaboration. The proposed an Android-based marketplace. This is designed to assist users in renting books, electronics, and other small furniture. Their work showed that user ratings and reviews make peer-to-peer exchanges trustable. The author [21-22], explored various models of rental platforms to identify the key challenges developers face. Intelligent Instrument Interface, a tool that allows music students and teachers to connect directly through peer-to-peer technology. This idea shows how similar systems can be used in universities for sharing or renting academic items. Together, these studies show that secure and user-friendly platforms are becoming more important. Although there is not much research about specific university rental platforms, a number of studies explore the solutions for academic communities. The author [23-25] suggested the Smart Student Rental Platform with the aim of maintaining the affordability and sustainability of living for students.

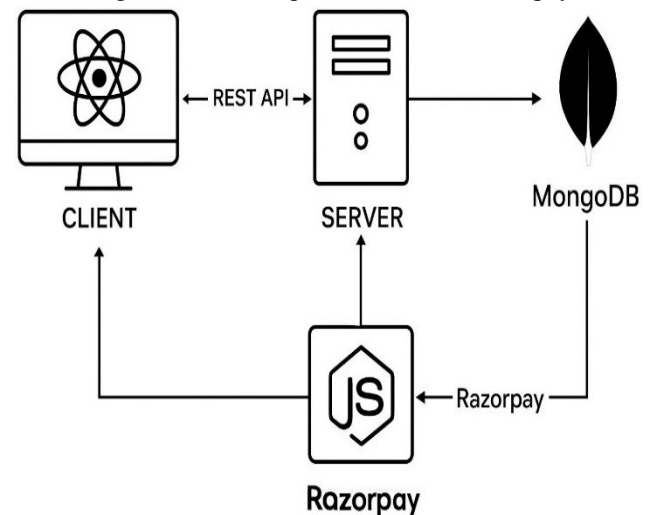
III. METHODOLOGY

Students are running into a lot of problems when renting things on or near campus. First of all, the rentals are not centralized in one place. Informal groups, as well as online social media and even community notice boards, are among them. It takes a lot of time and work to figure out what they need. Campus Rent removes these barriers and creates a safe online community for students. It pools all the types of rentals together [26-28].

System Design and Architecture:

Campus Rent is designed as a simple, three-piece web system: the part that the students see and use; the server that works in the background to get the job done and the database where all of the information is stored.

Client Layer (Frontend): The student UI is a web portal built in React that works great across phones and laptops. Although this interface, in which a user logs on, looks at items by category (books, lab equipment or the like), checks if something's available, requests it to borrow and pays for it



all in the one operation [29-30]. High level architecture (as shown in figure 1 above) is client browsers talks to Node/Express server and the server talks to MongoDB for data and Razor pay for payments.

Fig. 1. System Architecture of Campus Rent [30]

Server Layer (Back-end): The student UI is a React web frontend. This is where the rubber meets the road. It uses a Node. JS server with the Express framework for serving the frontend and a database. The server has routes for signing up, logging in, finding an item, scheduling an appointment and checking out payment. Because Node. js, a single thread can support many thousands of input- output connections, and not having to start a new thread for each concurrent student is very beneficial since the server can efficiently utilise the Database Layer [31-32].

Database (Data Layer): It is utilized as a repository by MongoDB to persist records for users, postings, reservation requests and transactions. Its "schema-less" document model applies to the differing ways in which items are organized by category. Schema definitions and queries are performed through Mongoose ODM [33-34].

A. Module Functionality:

The system comprises the following key modules as shown in Fig. 2:

Authentication: User registration, login, password hashing, and session.

Item Catalog & Search: CRUD operations for item listings; browse by category or search by keywords.

Booking: Create rental requests, select data range and confirm.

Payment: Integrate Razor pay to process transactions securely.

Administration: Dashboards for monitoring users, items, and rental statistics.

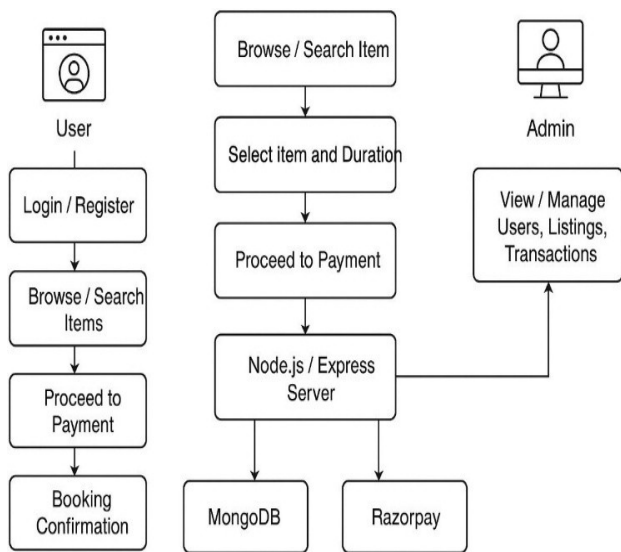


Fig. 2. Module Functionality

IV. PROPOSED SYSTEM ARCHITECTURE

This section discussed how each module was implemented using modern web technologies:

Authentication Module: Students register with their university email address and a password. Passwords are hashed-preferably with crypt and JSON Web Token is issued at login. MongoDB stores the user profiles with a unique student ID and email for verification [35].

Item Listing Module: Users can create a listing by giving information such as name, description, category, rental rate, condition, and images. Then, the server will store each listing in a MongoDB collection called Items. Users are allowed to make changes or delete their own items [36].

Category Browsing and Search: At the frontend, items are categorized into predefined categories, including Electronics, Tools, Sports, Furniture, Vehicles, Others. The various routes in Express query MongoDB for items that match with category or search terms; pagination for large result sets is also used [37].

Booking Module: Upon selecting an item, a renter specifies the dates/duration for which renting the item. The frontend sends this request to the server to create a provisional booking record. No capture of payment yet [38].

Payment Module: The server incorporates Razor pay's Node.js SDK. A Razor pay order is created for every booking by the server, which then responds with the ID of that order to the client. The frontend opens the Razor pay checkout. Once the payment is made successfully, Razor pay notifies the server through webhook. The server verifies the signature of the payment and marks the booking as confirmed in MongoDB [39].

Admin Dashboard: A special admin account can log in to a dashboard page. It displays real-time tables of all users, all listed items, and all completed transactions. Admins can also deactivate suspicious listings or users. Data is queried from MongoDB and displayed using simple HTML tables and

Figures 2. Through its implementation, the client and server communicated via a RESTful JSON API. Validation of incoming data occurs at the front-end form and backend route levels to prevent invalid data. For example, all required fields in a booking - dates, and item ID are checked on the server side to ensure consistent state.

TABLE 1. SYSTEM MODULES AND FUNCTIONALITIES

Module	Description
Authentication	Handles user registration, login, password encryption, and session management for secure access.
Item Catalog & Search	Allows CRUD operations for item listings, including browsing by category or keyword search.
Booking	Enables users to select items, choose rental duration, and create booking requests.
Payment	Integrates Razor pay gateway for secure online transactions through UPI, cards, or net banking.
Administration	Provides admin dashboard for managing users, listings, transactions, and system statistics.

V. RESULT & DISCUSSION

Functional correctness, time-constrained speed and a few Ux aspects of the system were checked. The test confirmations are: module-core: Account-registration-and-login, item-listing, booking-process-and-payment is good. In the test scenario, there was a whole booking test from login to confirmation of payment in approximately two minutes and author had a pretty good robust end-to-end pipeline. Performance-wise, Node.js (event-driven) back-end served concurrent requests very easily. The response time is in the scale of a few hundred milliseconds for systems that tens of thousands of users, as required by industrial rental platforms. In simulated testing, the Express.js server handled few hundreds of requests at the same time very well, keeping very similar latency even that a scalable web app for doing so. Future proposed experimentation author would like to conduct scaled-up stress testing so as to establish more thoroughly the system's robustness. A usage test with 20 students has revealed a strong acceptance in the user group. "Easy to navigate system, by category, with user friendly helping tool tips," wrote one rater about the user-friendly design of the booking feature. The integrated paywall and member verification served as a sort of perceived trust blanket. Believing in the platform was more acceptable for people seeking safety than ad hoc, peer-to-peer rental food swaps. In other words, the system architecture is tight and cheap in terms of costs and resources.

TABLE 2. RESULTS AND EVALUATION

Test Parameter	Observation	Remarks
Functional Testing	Core modules—including login, item posting, booking, and payments—worked smoothly during testing.	System functions verified successfully.

Performance	Average booking was completed under 2 minutes, and the platform handled over 200 simultaneous users.	Scalability targets achieved.
User Feedback	A pilot group of 20 students reported positive experiences highlighting ease of use and reliability.	Higher user satisfaction than traditional methods.
Security	User passwords were protected using encrypt, and all payments were securely processed through the Razor pay SDK.	Follows recommended security standards.
Usability	The interface was rated as easy to navigate and worked well on various devices.	Strong potential for widespread adoption.

VI. CONCLUSION

Campus Rent is made for those who are on campus, and offers a solution to rent & share everything among the members of this community. Renting has never been so simple through a well-managed tenant service and secure digital instant bad credit payday loans transfer processes. The author does it with ID-verified user accounts, so students can browse intuitively. The revised model is efficient because it encourages re-use and thus decreases waste by the shared use of these resources in academic environments. Over and above what already exists, Campus Rent has a high potential for development. An automatic-recommendation system might examine modules already rented, for instance, and recommend new ones to pick up or rent. IoT-enabled smart lockers or digital locks can also be included to facilitate hassle-free pickup and return, where the user may access their rental in a socially distanced manner. Expensive rentals might involve blockchain-powered smart contracts to ensure that nobody changes the terms of the lease or billing information. Right now, the platform operates via a web dashboard. But a new mobile app would make this process more user-friendly and accessible, with push notifications that are updated in real-time, better interaction between buildings around campus, and a unique user experience for each. Formal usability testing would more definitively hone this interface to fit students. Campus Rent is, in its essence, a brand-new way to fill the gap of campus services that lack serious competitiveness and trustworthy rental service in multiple categories within a close academic community. Beyond testing for systems' feasibility and usability, such an implementation of the setup could also identify cost savings.

REFERENCES

- [1] Kenneman, Johan, Daniel Lindén, Olof Höjeberg Lindstrand, August Ek, Mattias Nad, Fredrik Söderblom, Emil Bergkvist, and Elias Gustafsson. "Development and evaluation of a Trustworthy and User-Friendly bicycle rental platform for students." (2025).
- [2] Demchenko, Yuri, Juan J. Cuadrado-Gallego, Oleg Chertov, and Marharyta Aleksandrova. "Cloud Computing Foundation: Definition, Reference Architecture, Foundational Technologies, Use Cases." In *Big Data Infrastructure Technologies for Data Analytics: Scaling Data Science Applications for Continuous Growth*, pp. 81-113. Cham: Springer Nature Switzerland, 2024.
- [3] Kirschner, David Hans. "Empowering Community Connection: Building a Support Network Through Low-Code Product Development-A User-Centred Research Perspective." Master's thesis, Universidade NOVA de Lisboa (Portugal), 2023.
- [4] Nicula, Andrei-Sebastian, Andrei Ternauciuc, and Radu-Adrian Vasiiu. "A Smart Housing Recommender for Students in Timișoara: Reinforcement Learning and Geospatial Analytics in a Modern Application." *Applied Sciences* 15, no. 14 (2025): 7869.
- [5] Tripathi, Mukesh Kumar, D. N. Vasundhara, VKNSN Moorthy Ch, Kapil Misal, Bhagyashree Ashok Tingare, and Sanjeevkumar Angadi. "Hybrid deep learning system for crop disease classification using modified SegNet segmentation." *Computers and Electrical Engineering* 127 (2025): 110576.
- [6] Hao, Bo. "Research on Building a Web Full-Stack Development Talent Training System Based on Job Requirements." In *Proceedings of the 2025 International Conference on Educational Technology and Artificial Intelligence*, pp. 495-502. 2025.
- [7] Kumar, N. Suresh, Maddasani Mahidhar, Konduru Jay Charan, Chindukuri Ranjith Kumar, and Raavi Venkata Narayana. "A Full-Stack AI-based Multi-Stage E-Book Generator Leveraging Large Language Models Via Cohere AI." In *2025 7th International Conference on Innovative Data Communication Technologies and Application (ICIDCA)*, pp. 1475-1482. IEEE, 2025.
- [8] Marimuthu, Kavitha, Arunkumar Panneerselvam, Senthilkumar Selvaraj, Lakshmi Praba Venkatesan, and Vetrivelvi Sivaganesan. "Android based college app using flutter Dart." *Green Intelligent Systems and Applications* 3, no. 2 (2023): 69-85.
- [9] Singh, Gurjeet, Madiha Javed, and Balwinder Kaur Dhaliwal. "Full stack web development: Vision challenges and future scope." *International Research Journal of Engineering and Technology (IRJET)* 9, no. 04 (2022).
- [10] Brahmabhatt, Pratik, Shreyaskumar Patel, Mukesh Patidar, Daxa Vekariya, and Rashid Sheikh. "Green Cart: A Comprehensive E-Commerce Framework for Selling Plants and Gardening Services Online." In *2025 IEEE International Conference on Advances in Computing Research On Science Engineering and Technology (ACROSET)*, pp. 1-5. IEEE, 2025.
- [11] Alariksson, Daniel, Carolina Löfgren, Yra Olsen Öbrink, and Sebastian Rendel. "Sustainable Living: A digital solution for promoting social and ecological sustainability in student corridors." (2025).
- [12] Hasitha, Amaravadi, Chintapenta Shreya Sree, M. Neelakantappa, and Mukesh Kumar Tripathi. "A Machine Learning Approach Framework for Real-Time Diabetic Monitoring and Decision Support." In *2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, pp. 521-526. IEEE, 2025.
- [13] Ruchitha, Gangishetty, Dasari Jyoshna, Gnana Sri Koushik Galla, Indukuri Sai Manikanta Varma, Ch Madhu Babu, and L. Pallavi. "BookGenius: Elevating reading exploration through full-stack mastery." In *2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT)*, vol. 1, pp. 443-448. IEEE, 2024.
- [14] Dunka, Bakwa, Edim Emmanuel, and Dantala O. Oyerinde. "Simplifying Web Application Development Using-Mean Stack Technologies." *International Journal of Latest Research in Engineering and Technology (IJLRET)* (2018).
- [15] Mamushiane, Lusani, Albert Lysko, Hlabishi Kobo, and Joyce Mwangama. "Deploying a stable 5G SA testbed using srsRAN and Open5GS: UE integration and troubleshooting towards network slicing." In *2023 International Conference on Artificial Intelligence, Big Data, Computing and Data Communication Systems (icABCD)*, pp. 1-10. IEEE, 2023.
- [16] Gurupriya, M., and R. Gayathri. "E-Commerce Platform for Clothing: A Full Stack Web Application." In *2025 7th International Conference on Intelligent Sustainable Systems (ICISS)*, pp. 1022-1027. IEEE, 2025.
- [17] Tripathi, Mukesh Kumar, Meka Ananya, V. N. L. N. Murthy, Tumma Shashank, Jale Subash Reddy, and Kaushal Attaluri. "Enhanced Breast Cancer Relapse Prediction using Ensemble Learning Techniques." In *2024 4th International Conference on Sustainable Expert Systems (ICES)*, pp. 918-923. IEEE, 2024.
- [18] Gowda, P. G. A. N., and A. N. Gowda. "Streamlining data handling with full-stack web applications using Java and AngularJS." *International Journal for Multidisciplinary Research (IJFMR)* 2, no. 1 (2020): 168-174.
- [19] Drofa, Denys. "Integrating Advanced API Solutions into Full-Stack Web and Mobile Applications to Optimise User Experience." *International Journal of Current Science Research and Review* 8, no. 05 (2025).

- [20] Tricomi, Giuseppe, Maurizio Giacobbe, Ilenia Ficili, Nicola Peditto, and Antonio Puliafito. "Smart city as cooperating smart areas: On the way of symbiotic cyber-physical systems environment." *Sensors* 24, no. 10 (2024): 3108.
- [21] Kader, Md Abdul, and Rizvi Ahmed. "A Comprehensive Review on Full-Stack Development in the Cloud Era: Trends, Tools, and Best Practices." *Scientia. Technology, Science and Society* 2, no. 11 (2025): 310-318.
- [22] Hou, Jing, Guang Chen, Zhijun Li, Wei He, Shangding Gu, Alois Knoll, and Changjun Jiang. "Hybrid residual multiexpert reinforcement learning for spatial scheduling of high-density parking lots." *IEEE transactions on cybernetics* 54, no. 5 (2023): 2771-2783.
- [23] Aldorasi, Enrico Maria, Remo Pareschi, and Francesco Salzano. "Creatichain: From Creation to Market." In *International Conference on Image Analysis and Processing*, pp. 347-358. Cham: Springer Nature Switzerland, 2023.
- [24] Sri Rangelakshmi, M., Mukesh Kumar Tripathi, Ramesh Karnati, and D. Ganesh. "Analyzing Abusive Language Detection Using BERT Models." In *EAI International Conference on Intelligent Systems with Applications in Communications, Computing and IoT*, pp. 283-291. Cham: Springer Nature Switzerland, 2024.
- [25] Pandey, Maulishree, and Tao Dong. "Blending accessibility in UI framework documentation to build awareness." In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*, pp. 1-12. 2023.
- [26] Mohan, C., Saef Thallal, Mudit Mittal, Ashwini Kumar, Sonia Pandey, and Arpit Goel. "Automated Retail Checkout System with RFID Scanning and Wireless Billing for Retail Stores." In *2025 7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT)*, pp. 1-7. IEEE, 2025.
- [27] Cui, Jialin, Fangtong Zhou, Chengyuan Liu, Qinjin Jia, Yang Song, and Edward Gehringer. "Utilizing the constrained k-means algorithm and pre-class github contribution statistics for forming student teams." In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*, pp. 569-575. 2024.
- [28] Ilufoye, Habeeb, Oluwatolani Vivian Akinrinoye, and Chinelo Harriet Okolo. "A Circular Business Model for Environmentally Responsible Growth in Retail Operations." *International Journal of Multidisciplinary Research and Growth Evaluation* 1, no. 3 (2023): 107-113.
- [29] Spatharakis, Dimitrios, Ioannis Dimolitsas, Giacomo Genovese, Ioannis Tzanettis, Nikos Filinis, Eleni Fotopoulou, Constantinos Vassilakis et al. "A lightweight software stack for iot interoperability within the computing continuum." In *2023 19th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, pp. 715-722. IEEE, 2023.
- [30] Vepsäläinen, Juho, Petri Vuorimaa, and Arto Hellas. "The Potential of Serverless Edge-powered Islands for Web Development." *Journal of Web Engineering* 24, no. 1 (2025): 1-38.
- [31] Serena, Luca, Stefano Ferretti, Moreno Marzolla, and Gabriele D'Angelo. "An NFT-Based Solution to Enhance Trust in Decentralized Marketplaces." In *2025 IEEE Symposium on Computers and Communications (ISCC)*, pp. 1-6. IEEE, 2025.
- [32] Akhmetov, Bakhytzhan, Valery Lakhno, Nurzhama Oshanova, Volodimir Malyukov, Zhuldyz Alimseitova, Aliza Turgnbayeva, Inna Malyukova, and Miroslav Lakhno. "The game model of investing in the academic cloud." *TELKOMNIKA (Telecommunication Computing Electronics and Control)* 22, no. 6 (2024): 1369-1381.
- [33] Kadam, Sandeep, Mukesh Kumar Tripathi, Chaitali Shewale, Priya Shelke, Pravin R. Futane, and Suruchi Dedgaonkar. "Unveiling mental health with machine learning and deep learning: Exploring applications and navigating challenges." *Multidisciplinary Science Journal* 7, no. 5 (2025): 2025250-2025250.
- [34] Tenhunen, Saara, Tomi Männistö, Petri Ihtola, Jami Kousa, and Matti Luukkainen. "Software startup within a university-producing industry-ready graduates." In *2023 IEEE/ACM 45th International Conference on Software Engineering: Software Engineering Education and Training (ICSE-SEET)*, pp. 82-94. IEEE, 2023.
- [35] Taivalsaari, Antero, Tommi Mikkonen, Cesare Pautasso, and Kari Systä. "Full stack is not what it used to be." In *International conference on web engineering*, pp. 363-371. Cham: Springer International Publishing, 2021.
- [36] Gurusamy, Ashokkumar, and Ikram Ahamed Mohamed. "The evolution of full stack development: trends and technologies shaping the future." *Journal of Knowledge Learning and Science Technology ISSN: 2959-6386 (online)* 1, no. 1 (2020): 100-108.
- [37] Kamau, Eunice, Teemu Myllynen, Anuoluwapo Collins, Gideon Opeyemi Babatunde, and Abidemi Adeleye Alabi. "Advances in full-stack development frameworks: a comprehensive review of security and compliance models." *International Journal of Computer Applications* 182, no. 45 (2023): 1-10.
- [38] Lavanya P, Venkata, Mukesh Kumar Tripathi, Sangeetha K, and Janjhyam Venkata Naga Ramesh. "Hyperspectral crop image classification via ensemble of classification model with optimal training." In *Web Intelligence*, vol. 22, no. 4, pp. 627-657. Sage UK: London, England: SAGE Publications, 2024.
- [39] Palaparthi, S., Rayudu, Babu, V., Selvam, Pulikanti, J., & Chinnavenkaiah, K. K. (2025, January). Research on the Benefits of Full-Stack Web Development (MERN) and Mobile App Development (Flutter) for Growing Small Businesses. In *Proceedings of Eighth International Conference on Information System Design and Intelligent Applications* (pp. 291-311). Singapore: Springer Nature Singapore.